AQA A Level Biology

Topic 4 Genetic information, variation and relationships between organisms

Model answer notes by @biologywitholivia

Topic	Understand	Memorise	Practise
4.1 DNA, genes and chromosomes			
4.2 DNA and protein synthesis			
4.3 Genetic diversity can arise as a result of mutation or during meiosis			
4.4 Genetic diversity and adaptation			
Required practical 6			
4.5 Species and taxonomy			
4.6 Biodiversity within a community			
4.7 Investigating diversity			

4.1 DNA, genes and chromosomes

Compare and contrast DNA in eukaryotic cells with DNA in prokaryotic cells

Similarities:

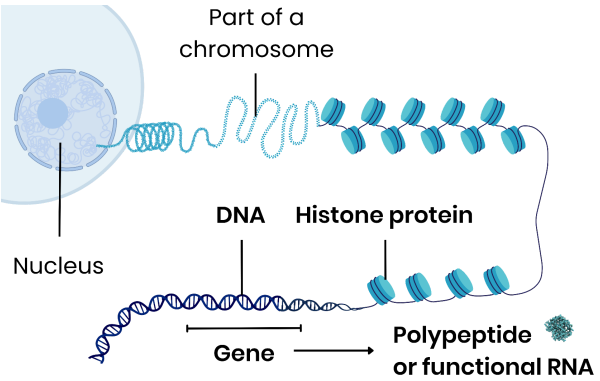
- **Nucleotide** structure is identical - deoxyribose attached to phosphate and a base
- Adjacent nucleotides joined by **phosphodiester bonds,** complementary bases joined by **hydrogen bonds**
- DNA in **mitochondria / chloroplasts** have similar structure to DNA in prokaryotes
    - **Short**, **circular**, **not** associated with **proteins**

Differences:

- Eukaryotic DNA is **longer**
- Eukaryotic DNA is **linear**, prokaryotic DNA is **circular**
- Eukaryotic DNA is associated with **histone proteins**, prokaryotic DNA is **not**
- Eukaryotic DNA contain **introns**, prokaryotic DNA does **not**

What is a chromosome?

- **Long, linear** DNA + its associated **histone proteins**
- In the **nucleus** of **eukaryotic** cells



What is a gene?

A sequence of **DNA (nucleotide) bases** that codes for:

- The **amino acid sequence** of a **polypeptide**
- Or a **functional RNA** (eg. ribosomal RNA or tRNA)

What is a locus?

Fixed position a **gene** occupies on a particular **DNA** molecule.

Describe the nature of the genetic code

Triplet code	A sequence of **3 DNA bases**, called a triplet, codes for a **specific amino acid**
Universal	The **same base triplets** code for the **same amino acids** in **all** organisms
Non-overlapping	Each base is part of only **one triplet** so each triplet is read as a **discrete** unit
Degenerate	An amino acid can be coded for by **more than one base triplet**

What are 'non-coding base sequences' and where are they found?

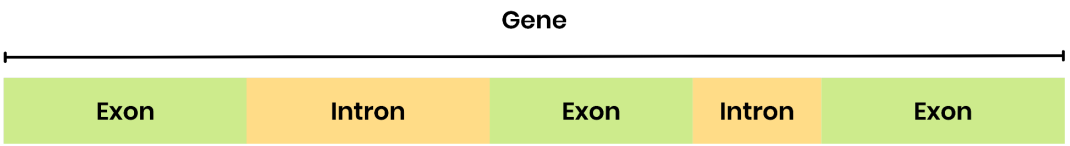
Non-coding base sequence - DNA that does **not code for amino acid sequences / polypeptides:**

1. **Between genes -** eg. non-coding multiple repeats
2. **Within genes** - introns

In **eukaryotes**, much of the **nuclear DNA** does not code for polypeptides.

What are introns and exons?

Exon	Base sequence of a gene **coding for amino acid sequences** (in a polypeptide)
Intron	Base sequence of a gene that **doesn't code for amino acids**, in eukaryotic cells



Exam insight: common mistakes ❌

Mistake	Explanation
"**Bacteria** / prokaryotic cells have **chromosomes**."	Chromosomes are found in eukaryotic cells. Bacteria have **small, circular DNA not associated with histones**.
"Prokaryotic DNA is **single stranded**, and contains **uracil** instead of **thymine**.	This describes **RNA**. Prokaryotic DNA is **double stranded** and made of exactly the same nucleotides as eukaryotic DNA (with A, T, C, G).
"Triplets **produce** amino acids."	Triplets **code** for amino acids, but don't directly produce them.
"A gene is a section of DNA that codes for **proteins**."	This is too vague. A gene is a **sequence of DNA (nucleotide) bases** that codes for a **polypeptide** (or functional RNA).
"An intron is a **non-coding** region of a gene."	This is correct, but **not enough** to get a mark. You must state what it is that introns don't code for, ie. **amino acids**.
"An exon codes for a **protein**."	This is too vague. An exon is a base sequence of a gene **coding for amino acid sequences** (in a polypeptide).

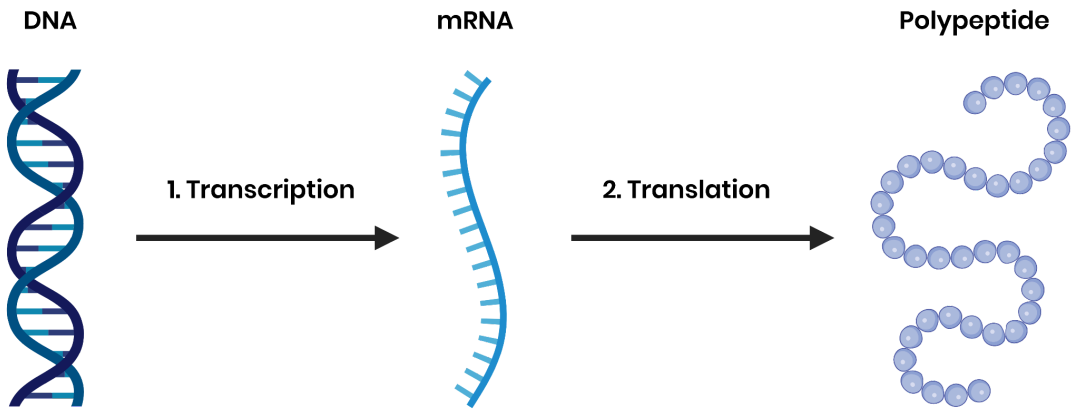
4.2 DNA and protein synthesis

Define 'genome' and 'proteome'

Genome	The complete set of **genes** in a cell (including those in mitochondria and /or chloroplasts)
Proteome	The full **range** of **proteins** that a cell can **produce** (coded for by the cell's DNA / genome)

Describe the two stages of protein synthesis

Transcription	Production of **messenger RNA (mRNA)** from DNA, in the **nucleus**
Translation	Production of **polypeptides** from the sequence of codons carried by mRNA, at **ribosomes**



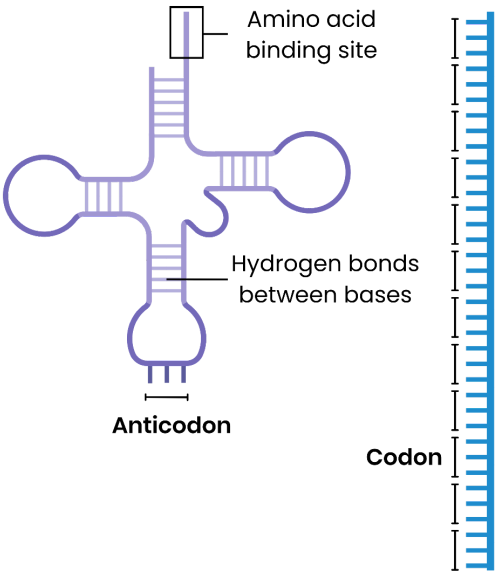
Compare and contrast the structure of tRNA and mRNA

Comparison (similarities)

- Both **single** polynucleotide strand

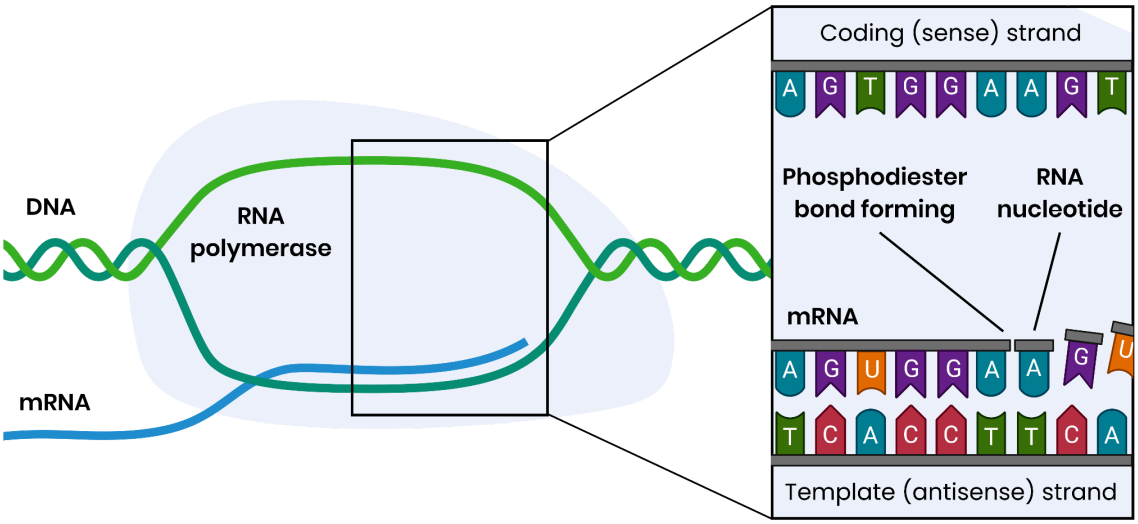
Contrast (differences)

- tRNA is folded into a **'clover leaf shape'**, whereas mRNA is **linear** / straight
- tRNA has **hydrogen bonds** between paired bases, mRNA doesn't
- tRNA is a **shorter, fixed** length, whereas mRNA is a **longer, variable** length (more nucleotides)
- tRNA has an **anticodon**, mRNA has **codons**
- tRNA has an **amino acid binding site**, mRNA doesn't



Describe how mRNA is formed by transcription in eukaryotic cells

1. **Hydrogen bonds** between DNA bases break
2. Only **one** DNA strand acts as a **template**
3. Free **RNA nucleotides** align next to their **complementary bases** on the template strand
   - In RNA, **uracil** is used in place of thymine (pairing with adenine in DNA)
4. **RNA polymerase** joins **adjacent** RNA nucleotides
5. This forms **phosphodiester bonds** via **condensation** reactions
6. **Pre-mRNA** is formed and this is **spliced** to **remove introns**, forming (mature) mRNA

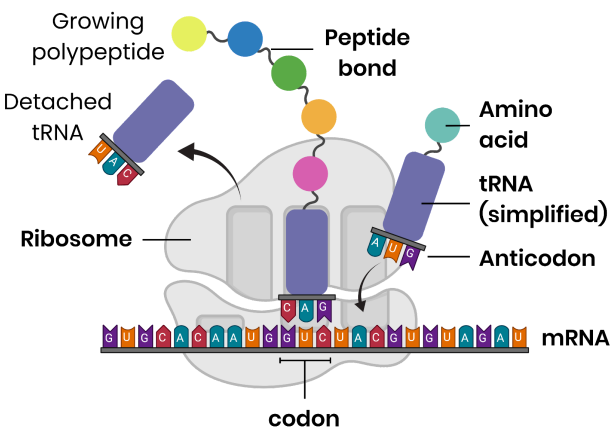


Describe how production of messenger RNA (mRNA) in a eukaryotic cell is different from the production of mRNA in a prokaryotic cell

- **Pre-mRNA** produced in **eukaryotic** cells whereas **mRNA** is produced directly in **prokaryotic** cells
- Because **genes** in prokaryotic cells **don't contain introns** so **no splicing** in prokaryotic cells

Describe how translation leads to the production of a polypeptide

1. mRNA attaches to a **ribosome** and the ribosome moves to a **start codon** (AUG)
2. tRNA brings a **specific amino acid**
3. **tRNA anticodon** binds to **complementary mRNA codon**
4. Ribosome moves along to **next codon** and another tRNA binds so **2 amino acids** can be joined by a **condensation reaction** forming a **peptide bond**
   - Using energy from **hydrolysis of ATP**
5. tRNA **released** after amino acid joined polypeptide
6. **Ribosome moves along** mRNA to form the polypeptide, until a **stop codon** is reached



Describe the role of ATP, tRNA and ribosomes in translation

ATP	• Hydrolysis of ATP to ADP + Pi **releases energy** • **So amino acids** join to **tRNAs** and **peptide bonds** form between amino acids
tRNA	• Attaches to / transports a **specific amino acid**, in relation to its anticodon • tRNA **anticodon complementary base pairs to mRNA codon**, forming hydrogen bonds • 2 tRNAs bring **amino acids** together so **peptide bond can** form
Ribosomes	• **mRNA** binds to ribosome, with space for 2 codons • Allows tRNA with **anticodons** to bind • Catalyses formation of **peptide bond** between amino acids (held by tRNA molecules) • **Moves along** (mRNA to the next codon) / translocation

Describe how the base sequence of nucleic acids can be related to the amino acid sequence of polypeptides when provided with suitable data

- You may be provided with a genetic code to identify which **triplets / codons** produce which **amino acids** (example shown)
- **tRNA anticodons** are **complementary** to **mRNA codons**
  - Eg. mRNA codon = ACG → tRNA anticodon = UGC
- Sequence of codons on **mRNA** are complementary to sequence of triplets on **DNA template** strand
  - Eg. mRNA base sequence = ACG UAG AAC
    → DNA base sequence = TGC ATC TTG
- In RNA, **uracil replaces thymine**

First base	Second base				Third base
	Guanine (G)	Adenine (A)	Cytosine (C)	Uracil (U)	
G	GGG Ala	GAG Glu	GCG Ala	GUG Val	G
	GGA Gly	GAA Glu	GCA Ala	GUA Val	A
	GGC Gly	GAC Asp	GCC Ala	GUC Val	C
	GGU Gly	GAU Asp	GCU Ala	GUU Val	U
A	AGG Arg	AAG Lys	ACG Thr	AUG Met	G
	AGA Arg	AAA Lys	ACA Thr	AUA Iso	A
	AGC Ser	AAC Asn	ACC Thr	AUC Iso	C
	AGU Ser	AAU Asn	ACU Thr	AUU Iso	U
C	CGG Arg	CAG Gln	CCG Pro	CUG Leu	G
	CGA Arg	CAA Gln	CCA Pro	CUA Leu	A
	CGC Arg	CAC Hist	CCC Pro	CUC Leu	C
	CGU Arg	CAU Hist	CCU Pro	CUU Leu	U
U	UGG Trp	UAG **stop**	UCG Ser	UUG Leu	G
	UGA **stop**	UAA **stop**	UCA Ser	UUA Leu	A
	UGC Cyst	UAC Tyr	UCC Ser	UUC Phe	C
	UGU Cyst	UAU Tyr	UCU Ser	UUU Phe	U

You should be able to **interpret data** from **experimental work** investigating the role of nucleic acids.

Exam insight: common mistakes ❌

Mistake	Explanation
"The genome is all the **genetic information** within a cell."	This is too vague. The genome is the complete set of **genes** or **DNA** in a cell.
"The genome is all the genes within a **species** or **population**."	The genome is the complete set of genes in a **cell**.
"The proteome is all the proteins **in a cell.**"	The proteome is the range of proteins that a cell can **produce.**
"tRNA carries / brings amino acids."	It does, but you need to say '**specific** amino acids' to get the mark.
"In transcription, polymerase joins nucleotides."	Be specific - **RNA** polymerase joins **RNA** nucleotides. Don't confuse this with DNA replication - DNA nucleotides and DNA polymerase.
"RNA polymerase forms **hydrogen bonds** / joins together **complementary bases**."	RNA polymerase joins **adjacent** (next to, not opposite) RNA nucleotides, forming **phosphodiester bonds**.

4.3 Genetic diversity can arise as a result of mutation or during meiosis

What is a gene mutation?

- A change in the **base sequence of DNA** (on chromosomes)
- Can arise **spontaneously** during **DNA replication** (interphase)

Examples - base deletion or substitution

What is a mutagenic agent?

A factor that **increases rate** of gene mutation, eg. ultraviolet (UV) light or alpha particles.

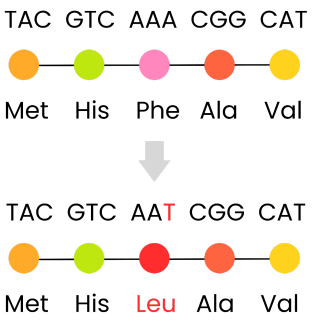
Explain how a mutation can lead to the production of a non-functional protein or enzyme

1. Changes sequence of **base triplets** in **DNA** (in a gene) so changes sequence of **codons** on **mRNA**
2. So changes sequence of **amino acids** in the polypeptide
3. So changes **position** of **hydrogen / ionic / disulphide bonds** (between amino acids)
4. So changes protein **tertiary** structure (shape) of protein
5. **Enzymes** - **active site** changes shape so substrate can't bind, **enzyme-substrate complex** can't form

Explain the possible effects of a substitution mutation

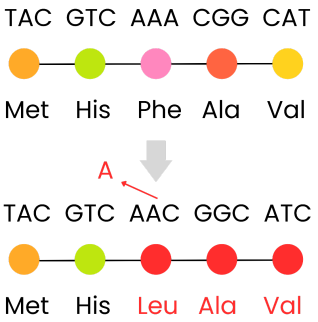
1. Base / nucleotide in DNA **replaced** by a different base / nucleotide
2. This changes **one triplet** so changes one **mRNA codon**
3. So **one amino acid** in polypeptide changes
   - **Tertiary** structure may change if **position** of hydrogen / ionic / disulphide bonds change

   OR amino acid **doesn't** change
   - Due to **degenerate** nature of genetic code (triplet could code for same amino acid) **OR** if mutation is in an **intron**



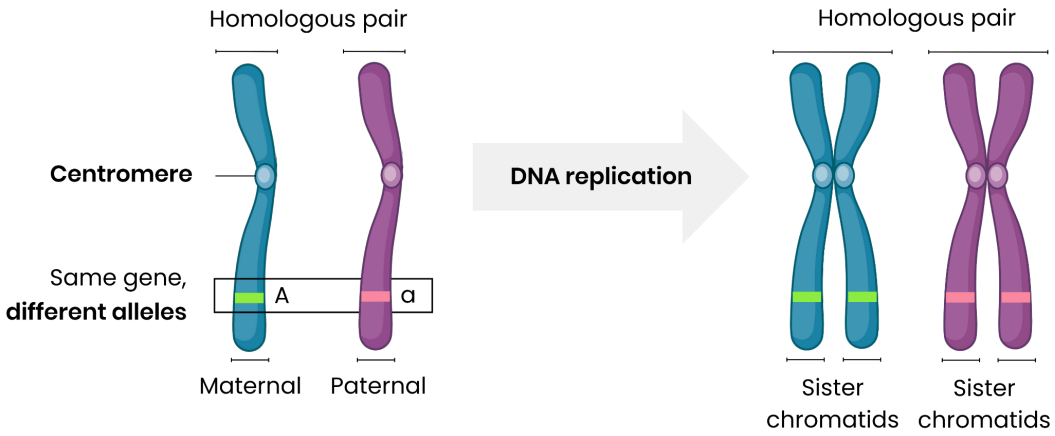
Explain the possible effects of a deletion mutation

1. One nucleotide / base **removed** from DNA sequence
2. Changes sequence of **DNA triplets from point of mutation** (frameshift)
3. Changes sequence of **mRNA codons** after point of mutation
4. Changes sequence of **amino acids** in primary structure of polypeptide
5. Changes position of **hydrogen / ionic / disulphide bonds** in tertiary structure of protein
6. Changes **tertiary structure** / shape of protein



Describe features of homologous chromosomes

Same **length**, same **genes** at same loci, but may have **different alleles**.



Describe the difference between diploid and haploid cells

- **Diploid** - has **2** complete sets of chromosomes, represented as 2n
- **Haploid** - has a **single** set of unpaired chromosomes, represented as n

Describe how a cell divides by meiosis

In **interphase**, **DNA replicates** → **2 copies** of each chromosome (sister chromatids), joined by a centromere.

1. **Meiosis I** (first nuclear division) separates **homologous chromosomes**
   - Chromosomes arrange into **homologous pairs**
   - **Crossing over** between homologous chromosomes
   - **Independent segregation** of homologous chromosomes
2. **Meiosis II** (second nuclear division) separates **chromatids**

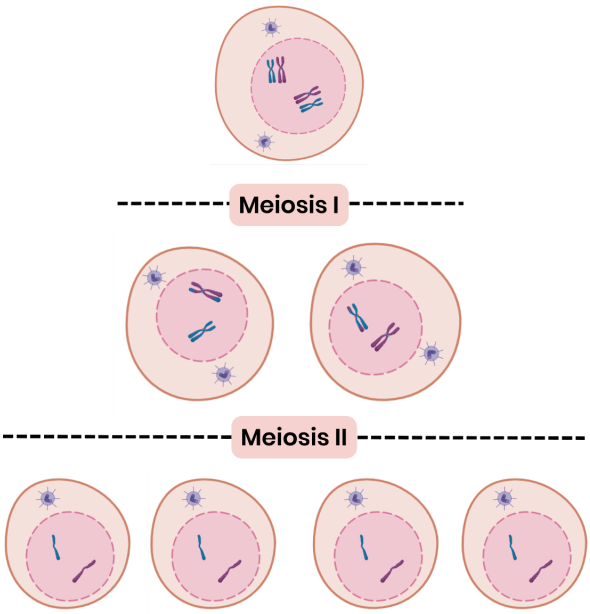
- Outcome = **4 genetically varied** daughter cells
- Daughter cells are normally **haploid** (if diploid parent cell)

Draw a diagram to show the chromosome content of cells during meiosis

You should be able to complete diagrams showing the **chromosome contents** of cells after the first and second meiotic division, when given the **chromosome content** of the **parent cell**.

In the example shown:

- Parent cell has **4 chromosomes** = **2 homologous pairs**
- These appear as X shapes due to **DNA replication**



Explain why the number of chromosomes is halved during meiosis

Homologous chromosomes are separated during **meiosis I** (first division).

Explain how crossing over creates genetic variation

- **Homologous pairs** of chromosomes associate / form a **bivalent**
- **Chiasmata** form (point of contact between (non-sister) chromatids)
- **Alleles** / (equal) lengths of (non-sister) chromatids **exchanged** between chromosomes
- Creating **new combinations** of (maternal & paternal) **alleles** on chromosomes

Explain how independent segregation creates genetic variation

- Homologous pairs **randomly** align at **equator** → so random which chromosome from each pair goes into each daughter cell
- Creating **different combinations** of maternal & paternal chromosomes / alleles in daughter cells

Other than mutation and meiosis, explain how genetic variation within a species is increased

- **Random fertilisation / fusion of gametes**
- Creating new **allele combinations** / new maternal and paternal **chromosome combinations**

Explain the different outcomes of mitosis and meiosis

1. **Mitosis** produces **2 daughter cells**, whereas **meiosis** produces **4 daughter cells**
    - As **1 division** in mitosis, whereas **2 divisions** in meiosis
2. **Mitosis maintains** the chromosome number (eg. **diploid** → **diploid** or **haploid** → **haploid**) whereas **meiosis halves** the chromosome number (eg. **diploid** → **haploid**)
    - As **homologous chromosomes separate** in meiosis but not mitosis
3. **Mitosis** produces **genetically identical** daughter cells, whereas **meiosis** produces **genetically varied** daughter cells
    - As **crossing over** and **independent segregation** happen in meiosis but not mitosis

Explain the importance of meiosis

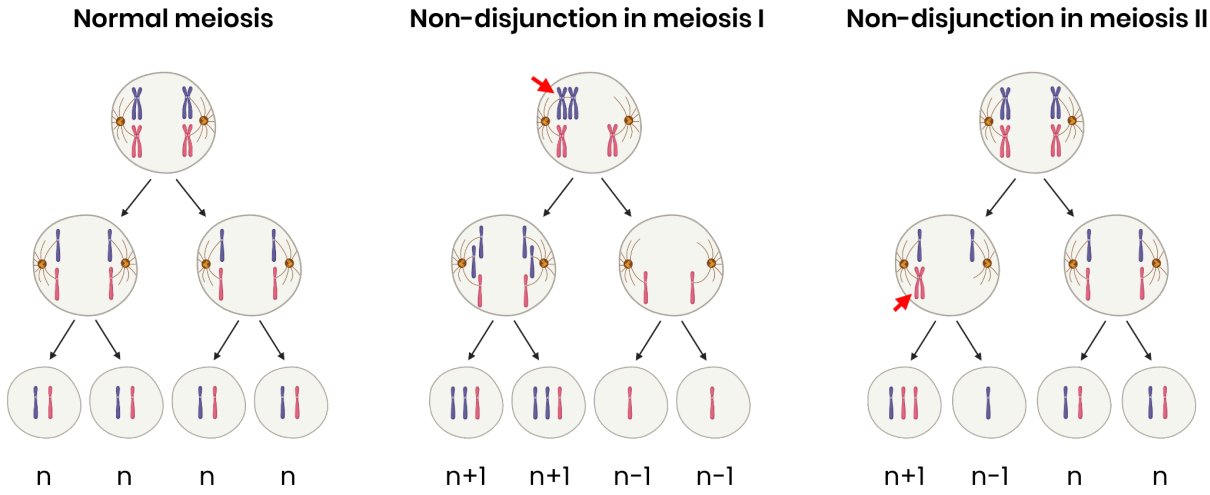
- **Two divisions** creates **haploid** gametes (halves number of chromosomes)
- So **diploid** number is restored at **fertilisation** → **chromosome number maintained** between generations
- **Independent segregation** and **crossing over** creates **genetic variation**

How can you recognise where meiosis and mitosis occur in a life cycle?

- **Mitosis** occurs between stages where chromosome number is **maintained** (eg. diploid (2n) → diploid (2n) OR haploid (n) → haploid (n))
- **Meiosis** occurs between stages where chromosome number **halves** (eg. diploid (2n) → haploid (n))

Describe how mutations in the number of chromosomes arise

- **Spontaneously** by **chromosome non-disjunction** during **meiosis**
- **Homologous chromosomes** (meiosis I) or **sister chromatids** (meiosis II) **fail to separate** during meiosis
- So some gametes have an extra copy (n+1) of a particular chromosome and others have none (n-1)



Suggest how the number of possible combinations of chromosomes in daughter cells following meiosis can be calculated

2^n where n = number of pairs of homologous chromosomes (half the diploid number)

Suggest how the number of possible combinations of chromosomes following random fertilisation of two gametes can be calculated

$(2^n)^2$ where n = number of pairs of homologous chromosomes (half the diploid number)

Exam insight: common mistakes ❌

Mistake	Explanation
"**All** mutations result in a change in amino acid sequence."	Due to the **degenerate** nature of the genetic code, many mutations do **not** affect the amino acid sequence of a polypeptide.
"Mutations change the **3D** structure of a protein."	Always mention the term **tertiary** structure and the change in the **position** of hydrogen bonds, ionic bonds and disulphide bridges.
"All mutations are **harmful**."	A mutation may result in the formation of a **new, advantageous allele** that results in change in polypeptide that positively affects the **properties** of the protein, resulting in a **selective advantage**.
"The **second** meiotic division halves the chromosome number."	Since the **first** meiotic division **separates homologous pairs**, it is actually this part that halves the chromosome number (eg. diploid → haploid).
"Crossing over creates **new alleles**."	Crossing over creates new **combinations** of (maternal & paternal) alleles on chromosomes .

4.4 Genetic diversity and adaptation

What is genetic diversity?

Number of different **alleles** of genes in a **population**

What are alleles and how do they arise?

- **Variations** of a particular **gene** (same locus) → different DNA **base sequence**
- Arise by **mutation**

What is a population?

A group of **interbreeding** individuals of the **same species.**

Explain the importance of genetic diversity

- Enables **natural selection** to occur
- As in certain environments, a new **allele** of a gene might **benefit** its possessor
- By resulting in a change in the **polypeptide** (protein) coded for that **positively** changes its **properties**
- Giving possessor a **selective advantage** (increased chances of survival and reproductive success)

What is evolution?

- **Change** in **allele frequency** (how common an allele is) over **many generations** in a population
- Occurring through the process of **natural selection**

Adaptation and **selection** are major factors in evolution and contribute to the **diversity** of living organisms.

Explain the principles of natural selection in the evolution of populations

1. Mutation	**Random gene mutations** can result in [named] **new alleles** of a gene
2. Advantage	In certain [named] **environments**, the new allele might **benefit** its possessor [explain why] → organism has a **selective advantage**
3. Reproductive success	Possessors are more likely to **survive** and have **increased reproductive success**
4. Inheritance	Advantageous allele is **inherited** by members of the next generation (**offspring**)
5. Allele frequency	Over **many generations**, [named] allele **increases** in **frequency** in the population

Natural selection results in species that are better adapted to their environment.

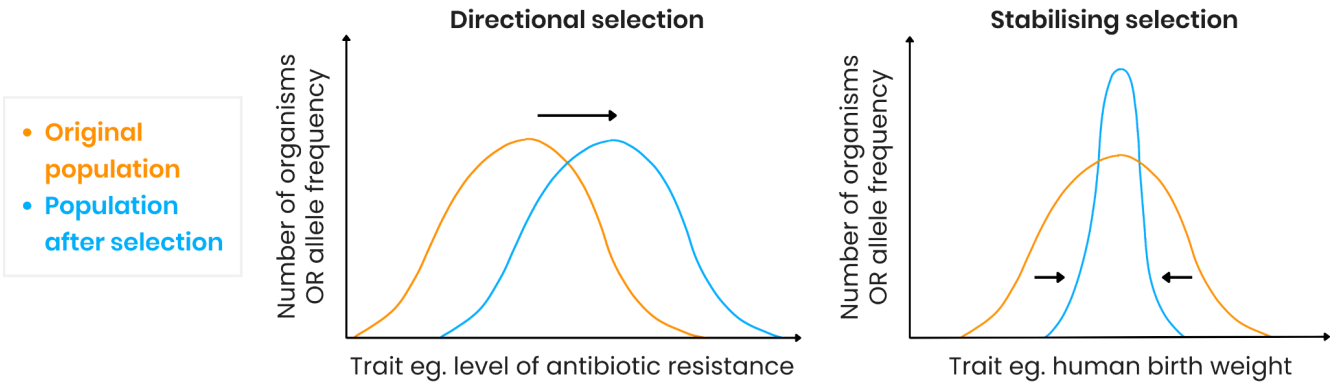
You should be able to use **unfamiliar information** to explain how selection produces **changes** within a population of a species and **interpret data** relating to the **effect** of selection in producing **change** within populations.

Describe 3 types of adaptations

- **Anatomical** - structural / physical features that increase chance of survival
- **Physiological** - processes / chemical reactions that increase chance of survival
- **Behavioural** - ways in which an organism acts that increase chance of survival

Explain two types of selection, with examples

	Directional selection	Stabilising selection
Example	**Antibiotic resistance** in bacteria	Human **birth weight**
Key feature - who has a selective advantage?	Organisms with an **extreme variation of a trait** eg. bacteria with high level of resistance to a particular antibiotic	Organisms with an **average / modal variation of a trait** eg. babies with an average weight
Change in environment?	**Yes**, usually eg. antibiotic introduced	**No**, usually **stable**
Effect on population over many generations	• **Increased** frequency of organisms with / alleles for **extreme trait** • Normal distribution curve **shifts** towards extreme trait	• **Increased** frequency of organisms with / alleles for **average trait** • Normal distribution curve similar, **less variation** around the mean



Exam insight: common mistakes ❌

Mistake	Explanation
"If I repeat the **generic answer** about natural selection, I'll get full marks."	Exam questions are almost always **application** based. Use information provided to **link** your answer to the question.
*Forgetting to mention **mutations** in an answer about natural selection.*	Mutations result in new **alleles** of genes, increasing **genetic diversity** and enabling natural selection to occur. This mark is commonly missed.
"Antibiotics **cause** mutations."	Mutations happen **continuously** and **randomly**. Exposure to antibiotics is the **selection pressure,** causing non-resistant bacteria to die and allowing resistant bacteria with the mutation to survive and reproduce.
"In antibiotic resistance bacteria become **immune** to antibiotics."	Bacteria **don't have immune systems.** Being resistant means they resist their effect, ie. they are not killed.

Required practical 6

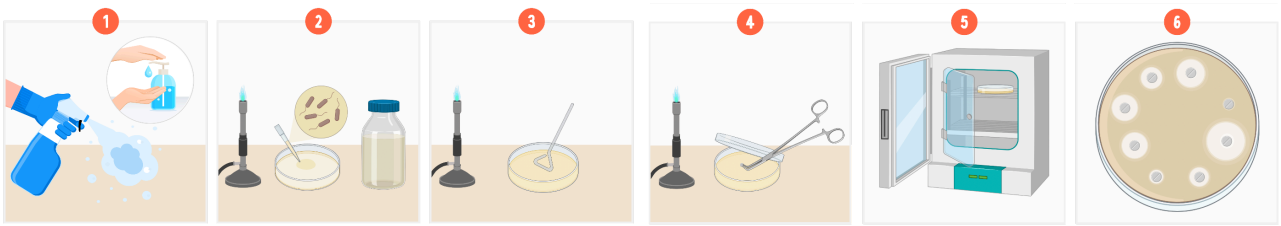
Use of **aseptic techniques** to investigate the **effect of antimicrobial substances** on microbial growth.

Explain examples of aseptic techniques that could be used

- Wash hands with **soap** / **disinfect** surfaces → **kill** microbes / prevent **contamination**
- **Sterilise pipette / spreader** / **boil agar** growth medium → **kill** microbes / prevent **contamination**
- **Flame neck** of bottle of bacteria → **kill** microbes / prevent **contamination**
- **Bunsen burner** close → **upward current** of air draws air-borne microbes away to prevent **contamination**
- Lift lid of petri dish **slightly** / minimise opening → **prevent entry** of microbes / **contamination**

Describe a method to investigate the effect of antimicrobial substances (eg. antibiotics, disinfectants, antiseptics) on microbial growth

1. Prepare area using **aseptic techniques** (as above)
2. Use a **sterile pipette** to transfer bacteria from **broth** to **agar plate** using **aseptic techniques** (as above)
3. Use a **sterile spreader** to evenly **spread** bacteria over agar plate
4. Use **sterile forceps** to place same size **discs** that have been soaked in different types / concentrations of **antimicrobials** for same length of **time,** onto agar plate (at equal distances)
5. **Lightly tape** lid onto plate (not fully sealed), **invert** and **incubate** at **25°C** for **48 hours**
6. Measure **diameter** of **inhibition zone** around each disc and calculate **area** using **πr^2**



Common questions:

Why is it important to maintain a pure culture of bacteria?	• Bacteria may **outcompete** bacteria being investigated • Or could be harmful to humans / pathogenic
Why hold lid with 2 pieces of tape instead of sealing it completely?	• Allows **oxygen** in preventing growth of **anaerobic** bacteria • Which are more likely to be **pathogenic / harmful to humans**
Why use a paper disc with water / no antimicrobial agent?	• Act as a **control** • Ensuring antimicrobial prevented growth, **not paper disc**
Why incubate upside down?	• **Condensation** drips onto lid rather than surface of agar
What if inhibition zones are irregular?	• **Repeat** readings in different **positions**, calculate a **mean**
Why not use higher antimicrobial conc.?	• More bacteria killed so clear zones may **overlap**
Why incubate at 25°C or less?	• Below **human body temp** to prevent growth of **pathogens**

Describe how data about the effect of antimicrobial substances can be presented as a graph

- **Categorical** data → **bar chart** (X axis type of antimicrobial, Y axis area of zone of inhibition / mm^3)
- **Continuous** data → **line graph** joined by a line of best fit (X axis concentration of antibiotic / μgmL^{-1}, Y axis area of zone of inhibition / mm^3)

Explain the presence and absence of clear zones

1. **Clear zones** → antimicrobial **diffuses** out of disc into agar, **killing / inhibiting growth** of bacteria
   - **Larger** clear zones → more bacteria killed → more effective antimicrobial
2. **No clear zones** → if antibiotic used, bacteria may be **resistant** or antibiotic may not be **effective** against that specific bacteria

4.5 Species and taxonomy

What is a species?

A group of organisms that can (interbreed to) produce **fertile offspring**

Suggest why 2 different species are unable to produce fertile offspring

- Different species have **different chromosome numbers** → offspring may have **odd chromosome number**
- So **homologous pairs** cannot form → **meiosis** cannot occur to produce gametes

Explain why courtship behaviour is a necessary precursor to successful mating

- Allows **recognition** of members of **same species** → so **fertile offspring** produced
- Allows recognition / attraction of **opposite sex**
- Stimulates / **synchronises mating** / production / release of **gametes**
- Indicates **sexual maturity** / fertility
- Establishes a **pair bond** to raise young

Describe a phylogenetic classification system

- Species (attempted to be) arranged into **groups**, called **taxa**, based on their **evolutionary origins** (common ancestors) and **relationships**
- Uses a **hierarchy**:
  - **Smaller groups** are placed within larger groups
  - **No overlap** between groups

Name the taxa in the hierarchy of classification

Domain (largest / broadest) → **kingdom** → **phylum** → **class** → **order** → **family** → **genus** → **species** (smallest)

How is each species universally identified?

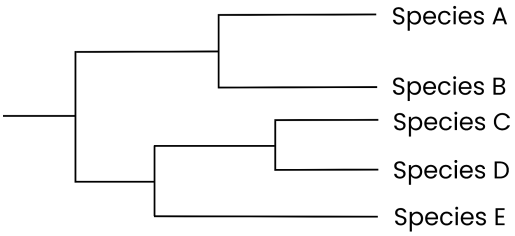
A **binomial** consisting of the name of its **genus** and **species**, eg. *Homo sapiens*

Suggest an advantage of binomial naming

Universal so no **confusion** as many organisms have more than one common name.

How can phylogenetic trees be interpreted?

- Branch point = **common ancestor**
- Branch = **evolutionary** path
- If two species have a more recent common ancestor, they are **more closely related** (eg. C & D)



Describe two advances that have helped to clarify evolutionary relationships between organisms

1. Advances in **genome sequencing** → allowing comparison of **DNA base sequences**
   - More **differences** in DNA base sequences → **more distantly related** / **earlier common ancestor**
   - As **mutations** (change in DNA base sequences) **build up over time**
2. Advances in **immunology** → allowing comparison of **protein tertiary structure** (eg. albumin)
   - **Higher amount** of protein from one species **binds** to **antibody** against the same protein from another species → **more closely related** / **more recent common ancestor**
   - As indicates a similar **amino acid sequence** and **tertiary structure**
   - So less time for **mutations** to build up

Exam insight: common mistakes ❌

Mistake	Explanation
"Two species are closely related if they share a **common ancestor**."	All species share a common ancestor. Two species are closely related if they share a **more recent** common ancestor.
"A **hierarchy** is smaller groups placed within larger groups."	The second part of this definition is that there is **no overlap** between groups. This part of the definition is often missed.
*Not being able to **interpret / draw circle diagrams** showing the relationship between the taxa in the classification of two different species.*	**Progressively smaller circles** are drawn within larger circles, each representing a taxa. For **shared** taxa, draw one circle. When taxa **diverge**, draw **non-overlapping** circles inside the last shared circle. Make sure you practise these types of questions.

4.6 Biodiversity within a community

What is biodiversity?

- **Variety** of living organisms (species, genetic and ecosystem diversity)
- Can relate to a range of **habitats**, from a small **local** habitat to the **Earth**

What is a community?

All **populations** of different species that live in an area.

What is species richness?

A measure of the **number** of **different species** in a **community**

What does an index of diversity do?

Describes the **relationship** between:

1. The **number of species** in a community (species richness)
2. The number of **individuals in each species** (population size)

Suggest why index of diversity is more useful than species richness

- Also takes into account **number of individuals in each species**
- So takes into account that some species may be present in **small** or **high numbers**

What is the formula for index of diversity?

$$d = \frac{N(N-1)}{\Sigma n(n-1)}$$

N = total number of organisms of **all species**

n = total number of organisms of **each species, Σ** = the **sum** of

List the steps involved in calculating an index of diversity

1. Calculate the total number of organisms (**N**), if not given
2. Multiply **N** by **(N - 1)**
3. For **each species**, multiply the number of organisms (**n**) by **(n - 1)**
4. Add up all the values of **n(n - 1)** to get **Σn(n - 1)**
5. **Divide** N(N - 1) by Σn(n - 1)

Describe how index of diversity values can be interpreted

- High → **many species** present (high species richness) and species **evenly represented**
- Low → habitat **dominated** by one / a few species

Index of diversity: worked example

Species	*Numbers in habitat X*	*Numbers in habitat Y*	*n(n-1) for X*	*n(n-1) for Y*
A	*10*	*3*	*90*	*6*
B	*10*	*5*	*90*	*20*
C	*10*	*2*	*90*	*2*
D	*10*	*36*	*90*	*1260*
E	*10*	*4*	*90*	*12*

1. *N for for both habitats - 50*
2. *N(N - 1) = 50 x 49 = 2450*
3. *n(n - 1) shown in additional columns of table*
4. *Σn(n - 1) for habitat X = 450 and for habitat Y = 1300*
5. *Habitat X = 2450 / 450 =* ***5.44***
   Habitat Y = 2450 / 1300 = ***1.88***

Explain how some farming techniques reduce biodiversity

Farming technique	Explanation
Removal of **woodland** and **hedgerows**	• Reduces **variety of plants** • So fewer **habitats** and niches • And less **variety of food sources**
Monoculture (growing one type of crop)	
Use of **herbicides** to **kill weeds**	
Use of **pesticides** to kill pests	**Predator** population of pest decreases

Explain the balance between conservation and farming

- Conservation required to **increase biodiversity**
- But when implemented on farms, **yields** can be reduced, reducing **profit** / **income** for farmers
  - Eg. by reducing land area for crop growth, increasing competition, increasing pest population
- To offset loss, **financial incentives / grants** are offered

Give examples of how biodiversity can be increased in areas of agriculture

- Reintroduction of **field margins** and **hedgerows** (where farmers only grow one type of crop)
- Reduce use of **pesticides**
- Growing **different crops** in the same area (intercropping)
- Using **crop rotation** of nitrogen fixing crops instead of fertilisers

Exam insight: common mistakes ❌

Mistake	Explanation
"Species richness is a measure of the number of different species in a **population**."	A population is a group of organisms of **one species**. Species richness is a measure of the number of different species in a **community / habitat / ecosystem / area**.
"Farming techniques reduce biodiversity as they result in **less food.**"	It's not that they cause less food overall, but that the **variety** of food sources decreases. This is vital to picking up the mark.

4.7 Investigating diversity

How can genetic diversity within or between species be measured?

- Comparing **frequency** of **measurable or observable characteristics**
- Comparing **base sequence** of **DNA**
- Comparing **base sequence** of **mRNA**
- Comparing **amino acid sequence** of a specific **protein** encoded by DNA and mRNA

Explain how comparing DNA, mRNA and amino acid sequences can indicate relationships between organisms within a species and between species

- More **differences** in sequences → **more distantly related** / **earlier common ancestor**
- As **mutations** (change in DNA base sequences) **build up over time**
- More mutations cause more changes in **amino acid sequences**

Explain the change in methods of investigating genetic diversity over time

- Early estimates made by inferring DNA differences from **measurable or observable characteristics**
  - Many **coded for by more than one gene** → difficult to distinguish one from another
  - Many influenced by **environment** → differences due to environment not genes
- Gene technologies allowed this to be replaced by direct investigation of **DNA sequences**

Explain the key considerations in quantitative investigations of variation within a species

- **Collect data** from **random** samples (use a random number generator) → **removes bias**
- Use **large** sample size (or sample until **stable running mean**) → **representative** of whole population
- **Ethical** sampling (where applicable) → must **not harm** organism / allow release **unchanged**
- **Calculate** a **mean** value of collected data and **standard deviation** of that mean
- **Interpret** mean values and their standard deviations (S.D)
  - S.D shows **spread of values about the mean** → higher S.D = higher variation
  - If standard deviations **overlap**, causing values of two sets of data to be **shared**, any difference between the two may be **due to chance / not significant**
- Use [named] **statistical test** → analyse whether there is a **significant difference** between populations

Exam insight: common mistake ❌

Mistake	Explanation
"**DNA** and **amino acids** measure genetic diversity within or between species."	This is too vague. **Comparing** DNA **base sequences** and amino acid **sequences** allows genetic diversity to be measured.

Created with BioRender.com